Strain2Force: Compact Nail-Based Sensor for 3DOF Contact Force Estimation

Yanjun Chen[1,†], Xiping Sun[2,†], Jung-Hwan Youn[1], and Craig Shultz[1],

***Abstract*—We present Strain2Force, a compact nail-mounted sensing system that continuously estimates fingertip contact forces through nail surface strain. The device integrates four strain gauges positioned at the four corners of the nail to capture localized strain variations. We examine placement strategies under normal and shear loading, corresponding to an average surface strain of approximately 0.02% per Newton of applied normal force. A machine learning regression model maps multichannel strain responses to three-dimensional fingertip forces, with high accuracy across all components (Fx, Fy, and Fz; R^2 = 0.877, 0.879, and 0.955; MSE = 0.028, 0.048, and 0.146). These results highlight a compact nail-based system capable of continuous, high-fidelity 3DOF force estimation through surface strain sensing.**

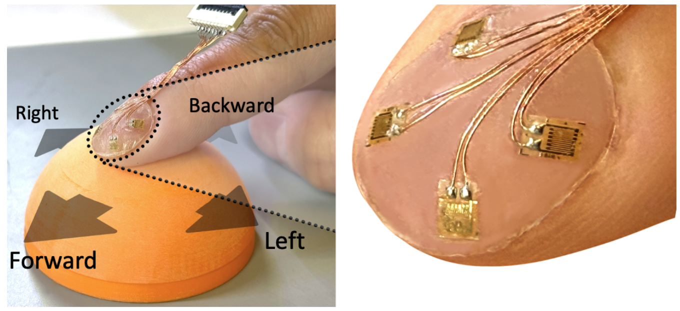

Fig. 1. Overview of the prototype device after adhered to a user's finger.

I. Introduction

The human finger is one of our most dexterous tools for perceiving and interacting with the physical world. We use it every day to sense, explore, and manipulate objects, from operating touchscreens to grasping small items. Understanding the interaction between the finger and objects reveals the mechanisms of human manipulation and provides a foundation for developing dexterous robotic hands [1] [2], effective teleoperation [3] [4], and rich haptic interactions [5]. However, accurately measuring contact forces on the finger is highly challenging. The contact forces are small and change rapidly, and their measurement is further complicated by soft-tissue deformation and nonuniform pressure distribution [6].

Vision-based methods can infer contact forces indirectly from external motion, but they are limited by spatial resolution and motion amplitude [7]. Meanwhile, tactile sensing provides a more direct means of measuring contact forces and deformations. Existing approaches span a wide range of modalities, including mechanical such as pressure [8], capacitive [9] and electric-field sensing [10], and barometric methods [11]. These methods require tradeoffs between sensing accuracy, practical wearability, and minimal interference with natural hand interaction [12]. This challenge has encouraged exploration of sensing strategies that can provide accurate and unobtrusive fingertip measurements.

Among these efforts, nail-based sensing has demonstrated strong potential for achieving compact and unobtrusive fingertip interfaces [13] [14] [15]. The fingernail is rigid and mechanically stable, providing a secure site for sensor attachment without affecting natural touch [9]. While fingerpad deformation is widely used for force estimation [16], this mechanical coupling extends to the nail, enabling nail-based sensing. Previous studies have demonstrated that strain signals measured on the nail surface can capture the small deformations [17] of the nail itself, which show some correlation with contact forces [18]. Patterns in strain over time have also been used to classify discrete fingertip motions such as normal pressing and shear interactions [19] [13].

Building on prior work in nail-based sensing, we present a tactile sensing system that measures fingertip contact forces through nail surface deformation. The system employs a compact artificial nail with four miniature strain gauges. We strategically place four gauges near the nail corners to capture deferential strain responses, a design choice guided by observed deformation sensitivity under contact. Additionally, we present a machine learning regression model, and show how it can be used to map these multichannel strain responses to continuous three-dimensional contact forces, providing estimation of both magnitude and direction with quantitative validation against ground-truth force measurements. This work demonstrates a compact nail-based approach for continuous estimation of 3DOF contact forces through surface strain sensing. Our aim is to advance nail based tactile sensing toward continuous, high fidelity force capture and to facilitate future research in robotic manipulation and haptic interaction.

II. Prototype Design

A. Sensing principle

The nail plate provides a relatively stiff surface that is well-coupled to the distal phalangeal bone. Consequently, when external forces are applied to the finger pad, the compliant

[†]Yanjun Chen and Xiping Sun contributed equally to this work.
[1]University of Illinois Urbana-Champaign, Urbana-Champaign, IL, USA (e-mail: yanjunc2@illinois.edu; jy92@illinois.edu; shultz88@illinois.edu).
[2]University of Pennsylvania, Philadelphia, PA, USA (e-mail: sxp22@seas.upenn.edu).

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fingertip pulp is compressed around the distal phalangeal bone and deforms the nail plate [13]. Through mechanical coupling with the nail bed, these deformations are transmitted to the nail plate, producing measurable micro-strains that are characteristic of fingertip loading. Under normal loading, compression of the fingertip pulp induces a somewhat uniform upward deflection of the nail plate. In contrast, lateral shear forces cause asymmetric bending and differential strain across the nail.

In this work, we measure subtle nail deformations using miniature strain gauges attached to a compliant artificial nail. We first establish the relationship between the applied fingertip forces and the resulting sensor signals, and then enable direct force prediction from the sensor outputs via a machine learning model.

B. Design and Implementation

Our prototype uses a compliant artificial nail as the base structure, with strain gauges mounted on its surface to detect subtle bending of the nail plate. This configuration enables a compact and flexible wearable device that can be securely attached to the user's fingernail. The prototype is compact and lightweight (0.3 g) and can be customized for different nail sizes. Each strain gauge (FPC type, model 120-1AA, Shenzhen Sensor and Control Co., Ltd., China) measures 3 mm × 2 mm and can respond to strains up to 20 000 µm/m. Its small footprint supports localized sensing and flexible placement on the artificial nail surface.

The overall structure and assembly of the prototype is shown in Fig. 1. The strain gauges were bonded to the ABS artificial nail using a medium-viscosity cyanoacrylate (instant) adhesive (3M PR1500). The prototype was then mounted onto the user's fingernail using cosmetic UV nail stickers intended for press-on nails. This adjesive is cured using 2-minutes of UV light exposure. This method provides strong mechanical coupling between the artificial and natural nails, enabling external forces applied on the fingerpad to be effectively transmitted to the strain gauges. To minimize cable-bending effects during finger motion, we used serpentine, compliant leads with slack routing. Finger flexion alone had negligible impact compared to contact loading.

We developed a high-precision multi-channel sensing circuit based on the ADS1263 10-channel 32-bit ADC module (Texas Instruments, USA), as shown in Fig. 2. Five input channels were used to record strain signals from the nail mounted gauges. Each channel connected a 120 Ω strain gauge in a quarter-bridge Wheatstone configuration with three matching precision resistors, via flexible flat cables (FFC), and the module interfaces with a Teensy 4.0 microcontroller through an SPI bus. Teensy collects digitized data from the ADS1263 and transfers them to a host PC via USB for real-time acquisition.

All circuits were powered from the 5 V USB supply of the Teensy 4.0, with a 3.3 V excitation applied to the ADS1263.

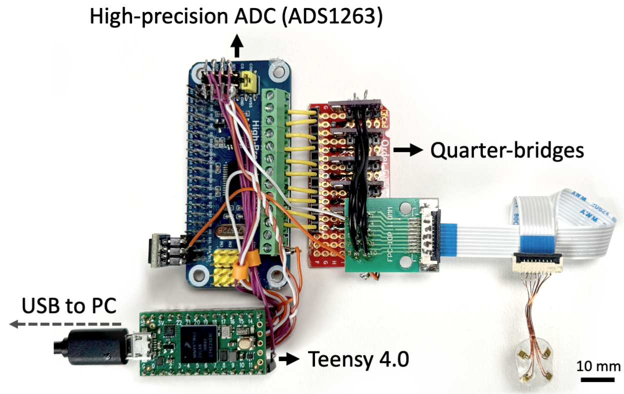

Fig. 2. Assembled electronics connected to the Strain2Force device.

The internal PGA (gain = 64) combined with digital filtering enhanced measurement resolution and suppressed noise. To convert the bridge output to physical strain, we used the standard quarter-bridge relation:

$$\varepsilon = \frac{\Delta L}{L} = \frac{4}{K\,V_{\mathrm{exc}}} V_{\mathrm{sig}}. \tag{1}$$

where $\varepsilon = \Delta L/L$ is engineering strain, L is the active gauge length (m), V_{sig} is the input-referred bridge output (V), V_{exc} is the bridge excitation (V), and K is the gauge factor.

After accounting for channel switching and settling, the effective sampling rate was about 315 Hz across up to five channels. To reduce long-term drift, we implemented an adaptive baseline algorithm that used signal variation to detect idle periods and slowly adjusted the baseline during these intervals.

C. Layout Design and Optimization

To investigate how strain distribution varies across different regions and orientations of the nail surface, we tested a layout configuration on the artificial nail with a 3×3 radial array (Fig. 3A). The goal was to analyze how local deformation responses correspond to fingertip loading and to determine layout patterns that better capture the deformation characteristics of the nail under applied normal and shear forces.

Normal and shear forces were recorded as a user pressed against the head of a high-precision six-axis force/torque sensor (ATI Nano17). These recordings served as a reference for measuring fingertip forces along the F_{x} (medial–lateral shear), F_{y} (proximal–distal shear), and F_{z} (dorsal–ventral, normal) directions. The participant performed five loading directions: press, left, right, forward, and backward, with 50 repetitions for each. The collected data were used to analyze strain distributions and evaluate how different regions of the nail responded to fingertip loading.

The nine strain gauges distributed across the nail surface exhibited strain changes ranging from -1.1% to 0.2%, corresponding to a maximum estimated surface elongation of

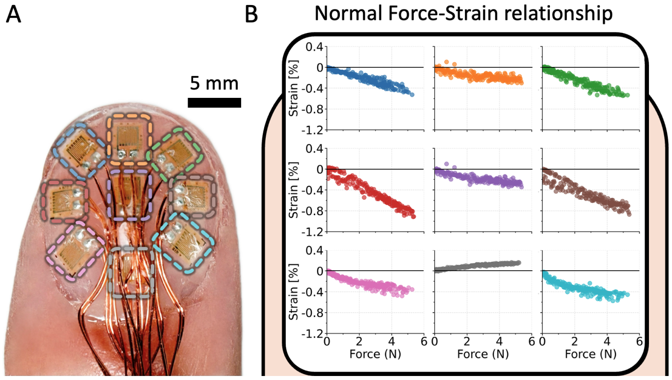

Fig. 3. A) Photograph of the strain gauge configuration on the artificial nail. B) Responses of the 3×3 strain gauge array under normal force loading, showing spatial variation across the nail surface.

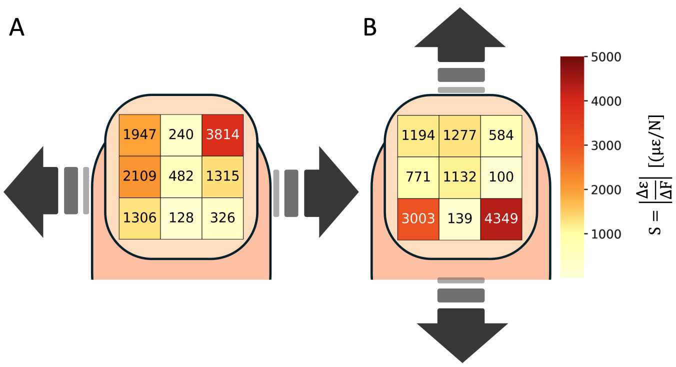

Fig. 4. Directional sensitivity maps showing absolute strain–force slope differences under (A) medial–lateral (left–right) and (B) proximal–distal (forward–backward) loading.

approximately 9 µm under a 5 N normal load, as shown in Fig. 3B. The responses showed a symmetric pattern between the left and right sides, with clear compressive strain at the lateral regions and weak deformation along the central axis, whereas the gauges near the nail base displayed opposite deformation. This is consistent with prior work, and the fact that nail strain should be reacted against through more rigid coupling of the underlying bone.

We further examined the relationship between lateral and longitudinal shear forces under corresponding left and right as well as forward and backward loading. Figure 4 shows that the four corner regions of the nail array exhibit consistently larger response differences across loading directions than the central regions, indicating that directional shear-induced strain is concentrated near the periphery. This observation suggests that many of the central gauges provide comparatively weak direction-informative strain signals for shear discrimination. Therefore, we reduced the sensing set to four corner elements, which capture the dominant direction-dependent strain components while minimizing channel count, wiring complexity, and calibration burden. This pattern is also consistent with previous Digital Image Correlation (DIC) measurements of

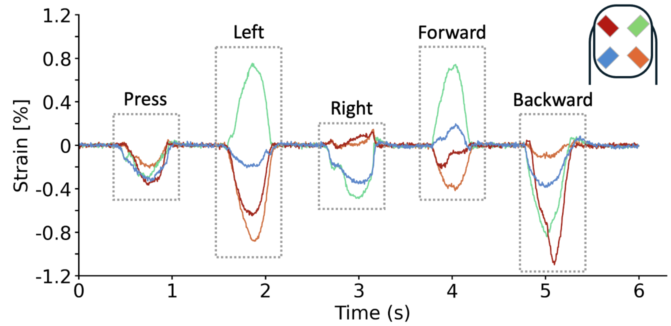

Fig. 5. Four strain signals showing distinct directional responses of four sensors during five fingertip motions.

lateral motion and shape changes reported in [20], and the observed directional trend provided guidance for the spatial arrangement of sensing elements in the subsequent prototype design.

III. Experiments

A. Sensor Input Configuration

In our final configuration, we aimed to detect fingertip gesture inputs and predict both normal and shear forces from the strain gauge signals using a regression-based approach. To achieve this, strain gauges were placed at the four corner regions of the fake nail. We hypothesize that this layout helps capture distinct strain responses under multi-directional loading, enabling accurate force estimation.

The strain gauge array demonstrated consistent and repeatable responses across multiple trials under both normal and shear loading conditions. Distinct signal patterns appeared during finger contact and release, with rapid fluctuations throughout loading and unloading. Under shear loading, the four strain gauges exhibited distinct response variations (see Figure 5). The fake nail structure provides mechanical stability and efficient strain transmission, enabling the strain gauges to capture unique signal variations across the nail surface and reveal characteristic strain patterns associated with each loading direction.

To estimate the direction and magnitude of fingertip forces in three degrees of freedom (3DOF), we developed a user-specific machine learning regression framework. This approach leverages the distinct strain distributions observed across the nail surface to learn user-calibrated mappings between strain signals and multi-axis forces. We assessed its performance through offline training and evaluation using limited samples.

B. Data Collection and Processing

We internally recruited five participants for data collection, and each participant performed four directional gestures (left, right, forward, and backward) under two force levels: light and heavy press. For each gesture–force combination, 30 trials were recorded, each lasting approximately 2 seconds. During recording, four strain gauges captured local deformation signals, while a 6DOF force–torque (FT) sensor simultaneously

TABLE I

PERFORMANCE METRICS FOR DIFFERENT FORCE DIRECTIONS AND STRAIN GAUGE CONFIGURATIONS. FORCES ARE IN NEWTONS (N).

Metric	**1 (Upper Right)**			**2 (Upper Corners)**			**4 (All Corners)**		
	F_x	F_y	F_z	F_x	F_y	F_z	F_x	F_y	F_z
MSE	0.182	0.382	0.825	0.059	0.148	0.631	0.028	0.048	0.146
MAE	0.143	0.207	0.341	0.098	0.147	0.284	0.064	0.083	0.149
R^2	0.189	0.039	0.745	0.737	0.629	0.805	0.877	0.879	0.955

measured ground-truth forces. For ground-truth measurement, all trials used contact on a rigid, planar surface mounted on the FT sensor, and participants maintained a controlled finger posture and fingertip orientation. These measurements produced synchronized multimodal data suitable for regression analysis. Given the broadly similar nail sizes in our sample, we used a single sensor geometry across all participants. The ABS nail base can be size-selected to accommodate different nail widths, and Fig. 3A shows representative fingertip fit and relative scale.

For model training, the data were segmented using a sliding-window approach (10-sample window, no overlap), enabling the model to capture short-term temporal dependencies in the strain responses. After segmentation, the dataset comprised 70 518 samples, each represented by four strain gauge inputs and three ground-truth force outputs (40-dimensional input per sample). This preprocessing preserved some temporal richness of the original recordings while producing a compact dataset suitable for faster learning and evaluation.

C. Machine Learning

After preprocessing, we trained a Multi-Layer Perceptron (MLP) regression model to predict three-dimensional forces from the strain gauge signals. The dataset was divided into 80% training and 20% validation subsets using a fixed random seed, which provided a consistent basis for model evaluation. Importantly, this split was performed at the window level rather than at the level of entire trials or recording sessions, meaning that windows originating from the same trial could appear in both subsets. This protocol was adopted to maximize the effective training data size given the limited number of recorded trials and participants. The network comprised three fully connected layers with ReLU activations and was optimized using Adam with a mean squared error (MSE) loss. Training employed early stopping based on validation loss to prevent overfitting. All implementations were developed in Python using TensorFlow.

D. Results

The performance of the MLP regression model is evaluated using mean squared error (MSE), mean absolute error (MAE), and the coefficient of determination (R^2). The model achieves high prediction accuracy across all three force components, demonstrating a strong correspondence between the measured strain signals and the fingertip forces. For the shear components, the model achieves R^2 values of 0.877 for F_x and 0.879 for F_y, indicating reliable estimation of lateral forces. The normal component F_z shows the best performance, with MSE = 0.146, MAE = 0.149, and an R^2 of 0.955, indicating the system's high sensitivity to normal loading. We further compared the proposed MLP regressor with several classical regression models, including Linear Regression (LR), Support Vector Regression (SVR), and Random Forest (RF). The results show that MLP substantially outperformed LR (R^2: 0.405, 0.413, 0.615 for F_x, F_y, and F_z) and SVR (R^2: 0.892, 0.86, 0.941 for F_x, F_y, and F_z) across all three force components, particularly for the normal force. While the RF model (R^2: 0.9, 0.897, 0.951 for F_x, F_y, and F_z) achieved slightly higher R^2 values for certain components, the overall performance of the MLP remained competitive, especially in terms of balanced accuracy across axes. These results confirm that combining four strain gauges substantially improves regression accuracy compared with single or dual gauge configurations (see Table I), validating the proposed multi-sensor design for capturing three-dimensional fingertip forces.

Figure 6A shows the predicted versus ground-truth forces for F_x, F_y, and F_z. In all three directions, the data points cluster closely along the diagonal red line, indicating that the model accurately estimated a linear mapping between strain gauge signals and measured forces. Among the three force components, predictions for F_z follows the ground truth most closely, while F_x and F_y exhibit slightly larger deviations, particularly at higher force magnitudes. This suggests that perpendicular deformation may be more reliably correlated with strain gauge readings, whereas shear forces are more difficult to regress due to their complex distribution across the sensor array.

To assess temporal generalization beyond the train/test split, we evaluate the model on a new single recording of 4 gestures from one participant that was not used in training or testing. The only preprocessing is the same non-overlapping 10-sample sliding window used elsewhere. Figure 6B shows representative sequences for F_x, F_y, and F_z. Across all three axes, the predicted trajectories closely track the ground-truth forces in both amplitude and timing, with clear agreement at gesture onsets/offsets and during sustained presses. Minor deviations appear primarily at sharp transitions and near the largest peaks (most visible in F_z), but the sign, peak order, and durations are preserved. These qualitative results indicate that the model maintains stable performance on unseen sequences with minimal preprocessing, supporting its suitability for real-time force estimation in interactive tasks.

E. Discussion

Although the random forest model achieved marginally higher regression accuracy in some cases, we selected the MLP as the final model due to its computational efficiency and suitability for real-time deployment. The MLP benefits from highly optimized GPU and CPU implementations in TensorFlow,

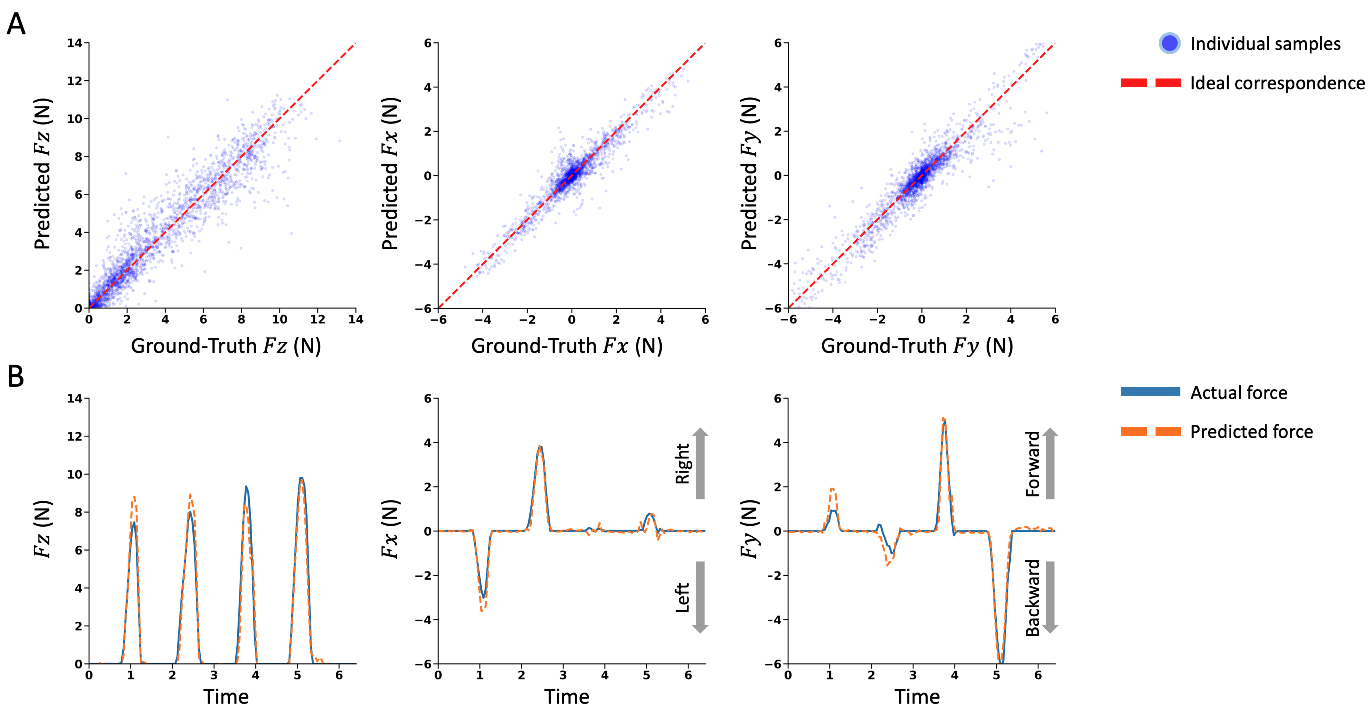

Fig. 6. Validation results showing (A) predicted versus ground-truth fingertip forces and (B) offline evaluation on unseen data to predict forces.

enabling fast inference with low latency, which is critical for continuous force estimation in interactive applications. In contrast, tree-based ensemble methods incur higher inference overhead and scale less favorably with increasing input dimensionality. Given the overall goal of enabling real-time force feedback from lightweight tactile sensors, the MLP provides a favorable trade-off between accuracy, speed, and deployability.

A limitation of the current evaluation protocol is that the train–test split was performed at the sliding-window level rather than at the level of entire trials or recording sessions. As a result, the reported metrics primarily reflect the model's ability to interpolate within the distribution of observed strain–force mappings, rather than its generalization to entirely unseen trials or participants. This choice was motivated by the limited size of the dataset and the need to ensure sufficient training samples for stable regression. While this approach is appropriate for assessing real-time force estimation performance under similar operating conditions, future work will adopt trial-level or subject-level splits and expand the dataset to more rigorously evaluate cross-trial and cross-user generalization.

These results highlight two additional important findings. First, the use of four strain gauges was critical for reliable force estimation. Modeling experiments including fewer sensors (only one or two gauges) failed to capture sufficient directional and spatial information, resulting in degraded regression performance, see Table I. Second, while the normal force predictions were highly accurate, the estimation of shear forces remains challenging, as evidenced by lower R^2 values. This suggests that incorporating additional sensing modalities or more advanced regression models may further improve shear force reconstruction. From our inspection of the strain gauge data, F_x appears to be sufficiently distinct for good regression, while F_y tends to be partially confused with F_z, likely due to the similar strain patterns produced during press and backward motions.

From a broader perspective, this study demonstrates the feasibility of combining low-cost strain gauges with machine learning regression for estimating three-degree-of-freedom forces. Such an approach could enable lightweight tactile sensors in human–robot interaction scenarios that require accurate, continuous force perception, or in human-computer interaction scenarios which could utilize subtle micro-gestures for wearable augmented reality (AR) input.

Future work could expand the dataset to improve generalization across users and tasks by recruiting a broader participant pool with diverse nail shapes and thicknesses, and by broadening data collection beyond the current controlled setup with planar rigid contact used for force ground truth to include a wider range of gestures, contact geometries, and finger postures. Another direction is to incorporate per-user calibration or adaptive learning methods to better personalize the model for individual differences, enabling the system to adjust to user-specific strain characteristics over time. In addition, integrating temporal modeling approaches such as recurrent or attention-based networks could help capture the dynamic evolution of strain–force relationships, while exploring multimodal sensor fusion may further enhance the estimation of complex shear forces.

IV. Conclusion

In this work, we presented a compact nail-based tactile sensing system that measures fingertip contact forces through nail deformation. The system integrates four strain gauges on an artificial nail and applies a machine learning regression model to map multichannel strain responses to continuous three-dimensional contact forces. Experimental results demonstrate that the four-sensor configuration enables reliable estimation of force magnitude and direction, achieving higher accuracy for normal forces than for shear forces. These findings validate the feasibility of continuous 3DOF force estimation from nail deformation signals and highlight the importance of sensor placement in multi-axis tactile sensing. The current strain gauge implementation can be affected by individual nail geometry and biomechanics and may risk mechanical failure after repeated use due to its delicate structure. Future iterations will explore using a single patterned strain gauge or alternative compact materials such as piezoelectric films (PZT or PVDF) to improve robustness and ease of wear. In future work, we plan to investigate how fingernails and multiple fingers interact with objects of different textures and shapes during grasping, and to analyze the resulting tactile signatures and force patterns to inform tactile sensing and perception in robotic manipulation.

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